SignVaani: PPT content, slide by slide

Content for the BMSCE mini-project PPT template (`MIP_PPT_template`), in the template's slide order. Each slide has what goes **on the slide**, the **figure** to insert, and **speaker notes** (what to say). Slides marked *optional* are extra slides you can add where shown.

Before the review:

- Replace every `[XX]` with a number you measured yourself. The command that produces each number is given on the Results slides. Never copy example numbers.
- The diagrams are in `docs/diagrams/`: use the `.png` files (Google Slides, any PowerPoint) or the `.svg` files (PowerPoint 2016 / 365 keeps them sharp).
- Reference numbers `[1]` to `[21]` match the References slides.

Slide 1: Title

ON THE SLIDE

- **Project title:** SignVaani: Real-Time Indian Sign Language to Speech
- Subtitle (smaller, optional): *A landmark-based, CPU-only system that turns ISL signs into English and Hindi speech*
- Name 1, USN 1
- Name 2, USN 2
- (Name 3, USN 3 and Name 4, USN 4 if you are a team of four)
- Guide: [Guide's name]
- Designation: [Assistant Professor / Associate Professor]
- Department of Computer Science & Engineering
- B.M.S. College of Engineering

SPEAKER NOTES

"Good morning. Our mini-project is SignVaani, which means 'voice of signs'. It recognises Indian Sign Language from an ordinary laptop webcam and speaks the meaning aloud in English or Hindi."

Slide 2: Outline

1. Introduction
2. Problem Statement and Objectives
3. Literature Survey
4. High Level Design
5. Detailed Design
6. Software and Hardware Requirements
7. Implementation
8. Results
9. Applications
10. Conclusion and Future Scope
11. References

Slide 3: Introduction

ON THE SLIDE

- Indian Sign Language (ISL) is the main language of India's deaf and hard-of-hearing community. It is a complete language with its own vocabulary and grammar.
- ISL is different from American Sign Language (ASL):
  - many signs use both hands;
  - facial expressions and head movements carry meaning (non-manual markers), e.g. raised eyebrows for a question;
  - sentences mostly follow subject-object-verb order, e.g. **ME STOMACH PAIN** = "My stomach hurts".
- The 2011 Census counted over **5 million** people in India with a hearing disability, but the country has only a few hundred certified ISL interpreters.
- So deaf people face daily communication barriers in hospitals, banks, railway stations and government offices, where staff do not know ISL.
- Most sign language recognition research and tools are built for ASL. ISL data and ready-to-use tools are scarce [18], [19].
- **SignVaani** recognises ISL signs from a normal laptop webcam, turns them into a grammatical sentence and speaks it in English or Hindi, in real time, on a laptop CPU, without a GPU or special sensor.
- Chosen domain: **hospital communication**, 15 signs + IDLE: ME, DOCTOR, PAIN, FEVER, HEAD, STOMACH, WATER, MEDICINE, TOILET, HELP, NEED, CALL, WHERE, YES, NO.

Figure: a simple flow across the bottom of the slide: `ISL sign -> Webcam -> Body/hand landmarks -> Sign recognition -> Sentence -> Speech`

Key terms (small box, optional)

Term	Meaning
Gloss	Written label of one sign, e.g. `PAIN`
Landmark	One (x, y, z) body point detected by MediaPipe
IDLE	The "no sign" class: resting hands, touching the face, adjusting hair
Signer-independent	The person tested never appears in the training data

SPEAKER NOTES

"ISL is not signed English. It has its own grammar, uses both hands and the face, and puts the verb at the end. Hospital staff rarely know it, and there are very few interpreters. Most research tools are for American Sign Language and need GPUs or depth cameras. We wanted something that runs on the laptop a hospital already has."

Slide 4: Problem Statement

ON THE SLIDE

PROBLEM STATEMENT

Deaf patients who use Indian Sign Language cannot easily communicate with hospital staff who do not know ISL, and certified interpreters are scarce. Existing sign language recognition systems mostly target ASL, need GPUs, depth cameras or gloves, stop at text output, and are usually evaluated on the same people they were trained on. There is a need for a low-cost system that recognises ISL signs from an ordinary webcam in real time, converts them into grammatical sentences, speaks them in English and Hindi, and works reliably for people it has never seen.

OBJECTIVES

1. Recognise 15 hospital ISL signs and an IDLE class from a laptop webcam in real time (at least 15 FPS) on a CPU.
2. Detect the start and end of each sign automatically, and ignore natural non-sign movements.
3. Convert the recognised signs (in ISL word order) into grammatical English and Hindi sentences and speak them.
4. Evaluate honestly with a signer-independent (leave-one-signer-out) protocol, and keep only landmarks, never video, for privacy.

SCOPE

In scope	Out of scope
Isolated recognition of 16 classes from a webcam	Continuous, open-vocabulary ISL translation
Automatic sign start/end detection	Fingerspelling A-Z
Rule-based ISL order to English / Hindi sentences	Neural translation, LLM sentence polishing
Speech: English offline (pyttsx3), Hindi via gTTS with offline cache	Fully offline Hindi TTS (future scope)
Signer-independent evaluation	Large user study or clinical validation
Own dataset: 5 signers x 16 classes x 10 takes = 800 clips	Mobile app

SPEAKER NOTES

"We deliberately fixed a small vocabulary so that every sign works on a new person. The golden rule we followed: 15 signs that work on someone the model has never seen are worth more than 50 signs that only work for the person who recorded them."

Slide 5: Literature Survey (1/4): ISL datasets and benchmarks

ON THE SLIDE (table)

No.	Paper (authors, venue, year)	Methodology / contribution	Limitations / gap
[1]	INCLUDE: A Large Scale Dataset for ISL Recognition (Sridhar et al., ACM Multimedia 2020)	Word-level ISL dataset: 4,287 videos, 263 signs in 15 categories, signed by deaf students. Best reported accuracy 94.5% on INCLUDE-50 and 85.6% on the full set	Lab dataset and recognition only; no live sentence building or speech output
[2]	CISLR: Corpus for Indian Sign Language Recognition (Joshi et al., EMNLP 2022)	Word-level ISL corpus with about 4,700 words; one-shot recognition baseline using features learned from ASL	One-shot recognition from video; no real-time sign spotting or speech pipeline
[3]	ISLTranslate: Dataset for Translating ISL (Joshi et al., Findings of ACL 2023)	About 31k ISL-English sentence pairs from educational videos; pose-based transformer translation baseline	Continuous translation accuracy is still low; heavy models; no speech output
[4]	iSign: A Benchmark for ISL Processing (Joshi et al., Findings of ACL 2024)	Large ISL-English benchmark (118k+ video-sentence pairs) with tasks such as SignVideo2Text, SignPose2Text and Text2Pose	A benchmark, not a deployable assistant; no domain focus or offline speech

SPEAKER NOTES

"ISL research has grown mainly through datasets. INCLUDE is the standard word-level dataset and we use its INCLUDE-50 subset to test our pipeline on real signing. But these works stop at recognition or translation scores; none gives a live, speaking assistant."

Slide 6: Literature Survey (2/4): ISL and multi-signer recognition

ON THE SLIDE (table)

No.	Paper (authors, venue, year)	Methodology / contribution	Limitations / gap
[5]	OpenHands: Pose-based Pretrained Models across Languages (Selvaraj et al., ACL 2022)	Uses pose keypoints instead of raw video; releases pose-based models for 6 sign languages including ISL, and a large ISL pretraining set	Research toolkit, not a real-time end-user app; no Indian-language speech
[6]	ISL-CSLTR: Dataset for Continuous Sign Language Translation and Recognition (Elakkiya and Natarajan, 2021)	Sentence-level ISL corpus: 700 videos of 100 sentences by 7 signers, plus word-level images	Small vocabulary and few signers; dataset only
[7]	Nearest Neighbour Classification of ISL Gestures using Kinect (Ansari and Harit, Sadhana 2016)	140 ISL signs from 18 subjects captured with a Kinect depth camera; k-NN classifier; about 90.7% on the fingerspelling subset	Needs a special depth camera; static gestures; no real-time sentences
[8]	Continuous SLR Handling Multiple Signers (Koller, Forster and Ney, CVIU 2015)	Statistical continuous recognition on RWTH-PHOENIX-Weather and SIGNUM; shows how hard multi-signer recognition is	German Sign Language; pre-deep-learning pipeline

SPEAKER NOTES

"OpenHands showed that pose keypoints work well for sign recognition, which is the approach we follow. Older ISL work used Kinect depth cameras, which a hospital does not have. Koller's work shows that accuracy drops when the signer changes, which is why we evaluate on unseen signers."

Slide 7: Literature Survey (3/4): large-scale recognition and translation

ON THE SLIDE (table)

No.	Paper (authors, venue, year)	Methodology / contribution	Limitations / gap
[9]	WLASL: Word-level Deep SLR from Video (Li et al., WACV 2020)	2,000+ ASL words from 100+ signers; compares appearance-based (I3D) and pose-based (Pose-TGCN) models	ASL only; models too heavy for real-time use on a laptop CPU
[10]	MS-ASL: Large-Scale Data Set and Benchmark for ASL (Vaezi Joze and Koller, BMVC 2019)	25,000+ videos of 1,000 ASL signs from 200+ signers; signer-independent splits; I3D baseline	ASL only; RGB video models need GPUs
[11]	Neural Sign Language Translation (Camgoz et al., CVPR 2018)	Treats sign language translation as neural machine translation; releases RWTH-PHOENIX-Weather 2014T	German Sign Language; offline research setting; no speech
[12]	Sign Language Transformers (Camgoz et al., CVPR 2020)	Transformer with CTC loss learns recognition and translation together; 21.80 BLEU-4 on PHOENIX14T	Large model; not ISL; no real-time deployment

SPEAKER NOTES

"The large ASL and German datasets drove the deep learning methods, but they use RGB video models that need GPUs. MS-ASL is important for us because it uses signer-independent splits, the same honest evaluation we use."

Slide 8: Literature Survey (4/4): pose and keypoint-based methods

ON THE SLIDE (table)

No.	Paper (authors, venue, year)	Methodology / contribution	Limitations / gap
[13]	SPOTER: Sign Pose-based Transformer (Bohacek and Hruz, WACV Workshops 2022)	Lightweight transformer on 2D body landmarks with normalisation and rotation augmentation; 63.18% on WLASL100	ASL / LSA64 only; no ISL, sentence or speech module
[14]	SAM-SLR: Skeleton Aware Multi-modal SLR (Jiang et al., CVPR Workshops 2021)	Skeleton graph network (SL-GCN) plus RGB and depth streams; won the 2021 signer-independent isolated SLR challenge (98.42% RGB track)	Heavy multi-modal ensemble; Turkish SL; not real-time on CPU
[15]	SignBERT: Hand-Model-Aware Pre-training (Hu et al., ICCV 2021)	Self-supervised pre-training on hand-pose tokens, then fine-tuning; state of the art on MS-ASL and WLASL	Hands only (no face or body); heavy pre-training; not ISL
[16]	ST-GCN: Spatial Temporal Graph Convolutional Networks (Yan, Xiong and Lin, AAAI 2018)	Graph convolution over skeleton joints in space and time; basis of many skeleton-based SLR models	Built for action recognition, not sign language; a possible future model for us

Tools used (one line under the table): MediaPipe [17] and MediaPipe Hands [18] give real-time pose, hand and face landmarks on a CPU.

SPEAKER NOTES

"Keypoint methods are compact, ignore the background and protect privacy, because no video is needed. SPOTER inspired our shoulder-based normalisation and rotation augmentation. SignBERT uses hands only; we add body and face points because ISL meaning depends on where the hand is and on facial expression."

Slide 9: Comparison of the existing methods

ON THE SLIDE (table)

Method	Sign language	Input	Reported result	Main limitation
Ansari and Harit [7]	ISL	Kinect depth camera	about 90.7% (fingerspelling subset)	Special sensor; static gestures
INCLUDE [1]	ISL	RGB video	94.5% (INCLUDE-50), 85.6% (263 signs)	No live sentences or speech
OpenHands [5]	ISL + 5 others	Pose keypoints	Pretrained pose models	Toolkit, not a real-time app
SPOTER [13]	ASL, LSA64	2D body landmarks	63.18% (WLASL100)	No ISL; no sentence or speech
SAM-SLR [14]	Turkish SL	Skeleton + RGB + depth	98.42% (RGB track)	Heavy ensemble; not real-time on CPU
Sign Language Transformers [12]	German SL	RGB video	21.80 BLEU-4 (PHOENIX14T)	Large model; no real-time use
SignVaani (ours)	**ISL (15 hospital signs + IDLE)**	**Laptop webcam, 63 hand, body and face landmarks**	**See Results**	**Small vocabulary of isolated signs**

RESEARCH GAP (box under the table)

None of the reviewed works combines all of the following, and SignVaani is designed to:

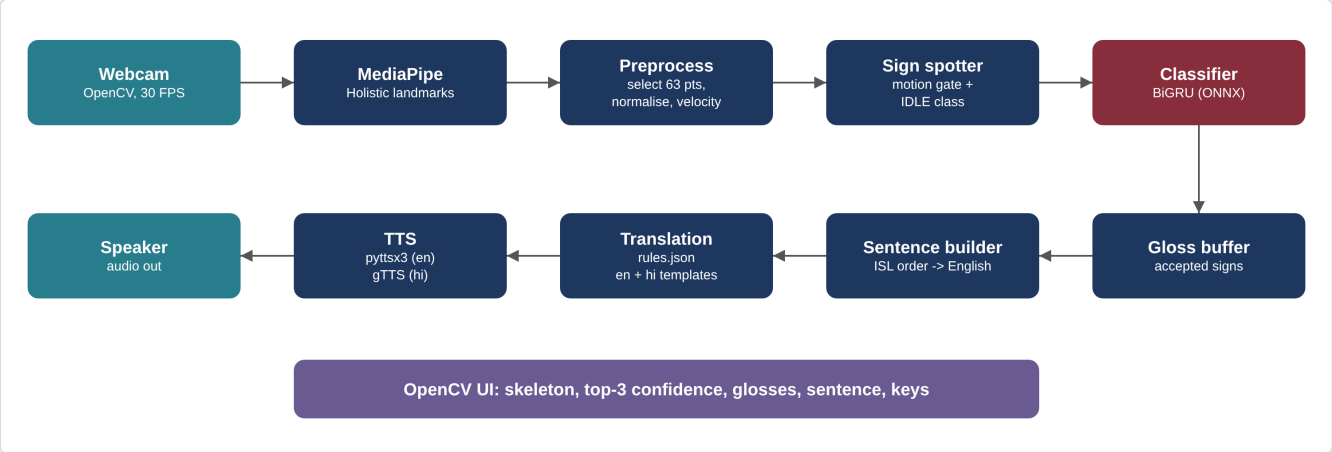
1. real-time ISL recognition on a laptop CPU;
2. hand + body + face landmarks;
3. automatic sign spotting with an IDLE class;
4. ISL-order to grammatical sentence generation;
5. speech output in an Indian language;
6. offline, privacy-preserving operation (no video stored);
7. signer-independent evaluation.

SPEAKER NOTES

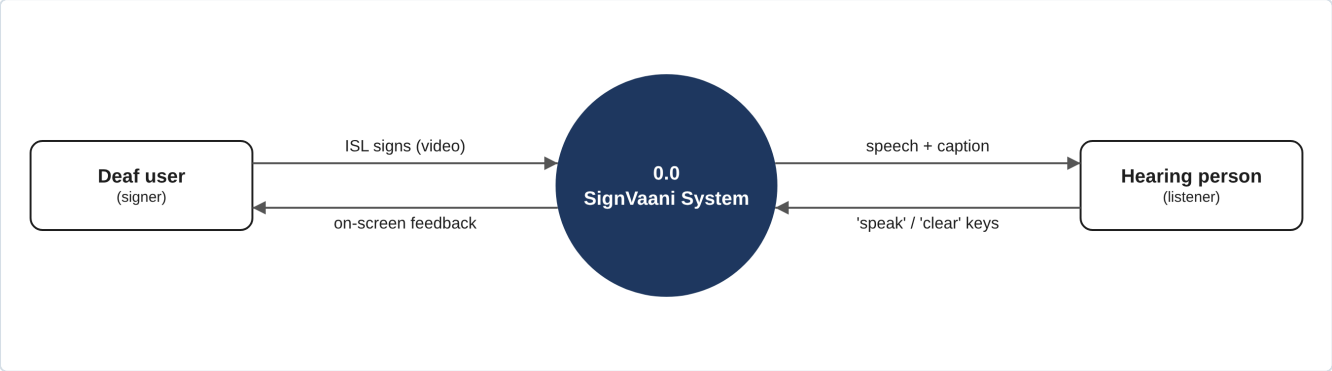
"The numbers in this table come from different datasets, so they are not directly comparable. The point is the last column: each work solves one part of the problem. Our contribution is putting the parts together into one working, honest system on cheap hardware."

Slide 10: High Level Design

Figure: `docs/diagrams/architecture.png` (system architecture). Optional on the same or the next slide: `docs/diagrams/dfd0.png` (context diagram, DFD level 0).



architecture.png



dfd0.png

ON THE SLIDE (one line per block, beside or under the figure)

Top row: runs on every frame (30 FPS)

- **Webcam:** OpenCV captures frames at 30 FPS.
- **MediaPipe Holistic:** finds pose, both-hand and face landmarks in each frame.
- **Preprocess:** keeps 63 points, fills short gaps, normalises to the shoulders, adds velocity.
- **Sign spotter:** hand-motion gate decides when a sign starts and ends; IDLE class rejects non-signs.
- **Classifier:** BiGRU with attention, exported to ONNX for fast CPU inference.

Bottom row: runs once per accepted sign or sentence

- **Gloss buffer:** list of accepted signs (undo / clear by keyboard).
- **Sentence builder:** ISL gloss order to a grammatical sentence (rules in `rules.json`).
- **Translation:** English and Hindi templates.
- **TTS:** pyttsx3 (English, offline), gTTS (Hindi, with an offline cache).
- **Speaker:** audio out; the caption stays on screen.

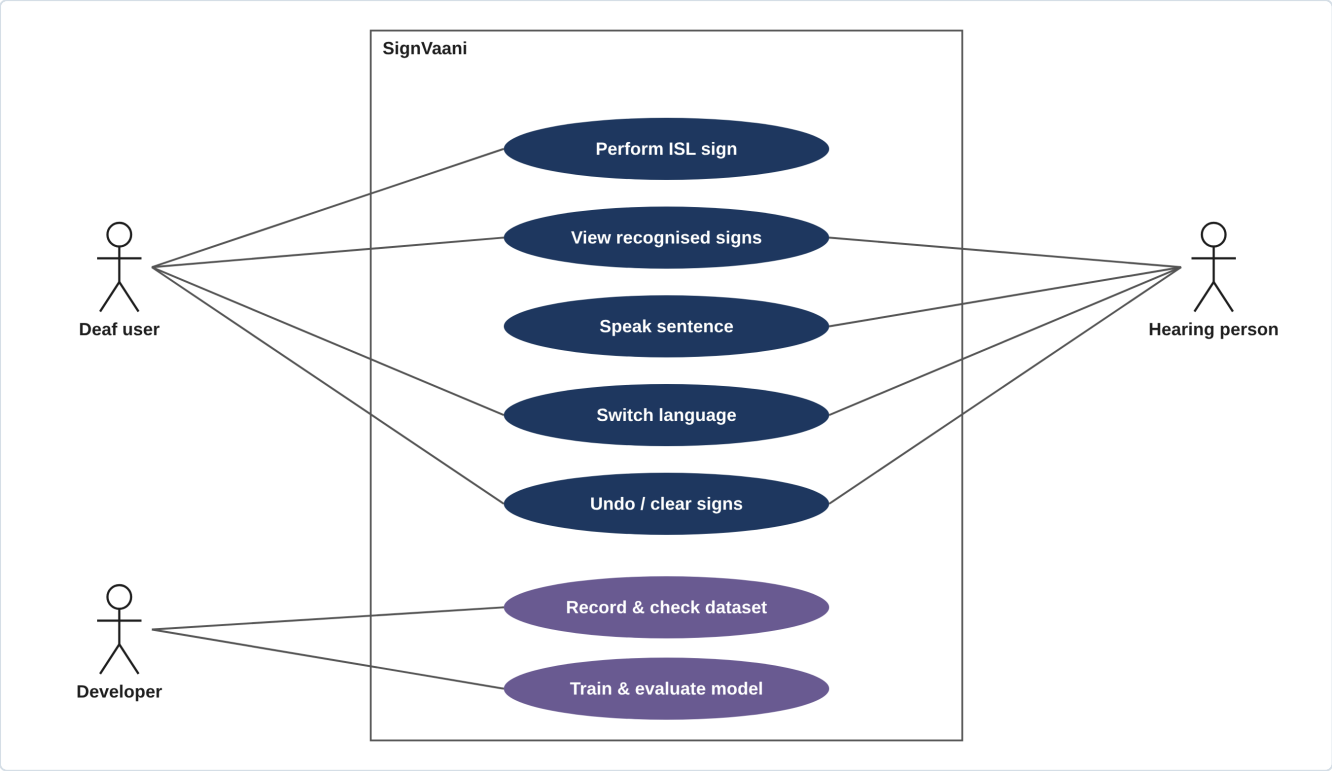
Design decisions (small box, optional)

- Landmarks instead of raw video: small input, background-independent, privacy-preserving.
- Explicit IDLE class + motion gate: no false words while the user is resting.
- Rule-based sentences: predictable, cannot invent content that was not signed.
- ONNX Runtime: fast inference on any laptop CPU.

> *SPEAKER NOTES*
>
> *"The top row runs 30 times a second. The bottom row runs only when a sign is accepted or the user asks for speech. Only landmark coordinates are kept, never video, which makes the system private and light."*

Optional extra slide after slide 10: High Level Design (contd.): Use cases

Figure: `docs/diagrams/usecase.png`.



usecase.png

ON THE SLIDE

Actor	What they do
Deaf user (signer), e.g. a patient	Signs in front of the webcam; sees the recognised signs and the sentence; undoes a wrong sign
Hearing person (listener), e.g. hospital staff	Hears the sentence and reads the caption; presses SPACE to speak or `c` to clear
Developer (project team)	Records signers, checks the data, trains, evaluates and exports models

Slide 11: Detailed Design

ON THE SLIDE

1. Feature vector (per frame)

- 63 landmarks per frame: 9 upper-body pose points (nose, eyes, shoulders, elbows, wrists) + 12 face points (lips, eyebrows, eye corners) + 21 left-hand + 21 right-hand points.
- 63 x (x, y, z) = 189 position values + 189 velocity values = **378 features per frame**.
- Each sign is resampled to 32 frames, so the model input is **32 x 378**.
- A body part that was not detected is stored as NaN (not zero); gaps of up to 6 frames are interpolated.

2. Normalisation (removes the effect of where the person stands and their body size)

- p' = (p - shoulder centre) / shoulder width
- v(t) = p'(t) - p'(t-1)

3. Classifier: BiGRU with attention pooling

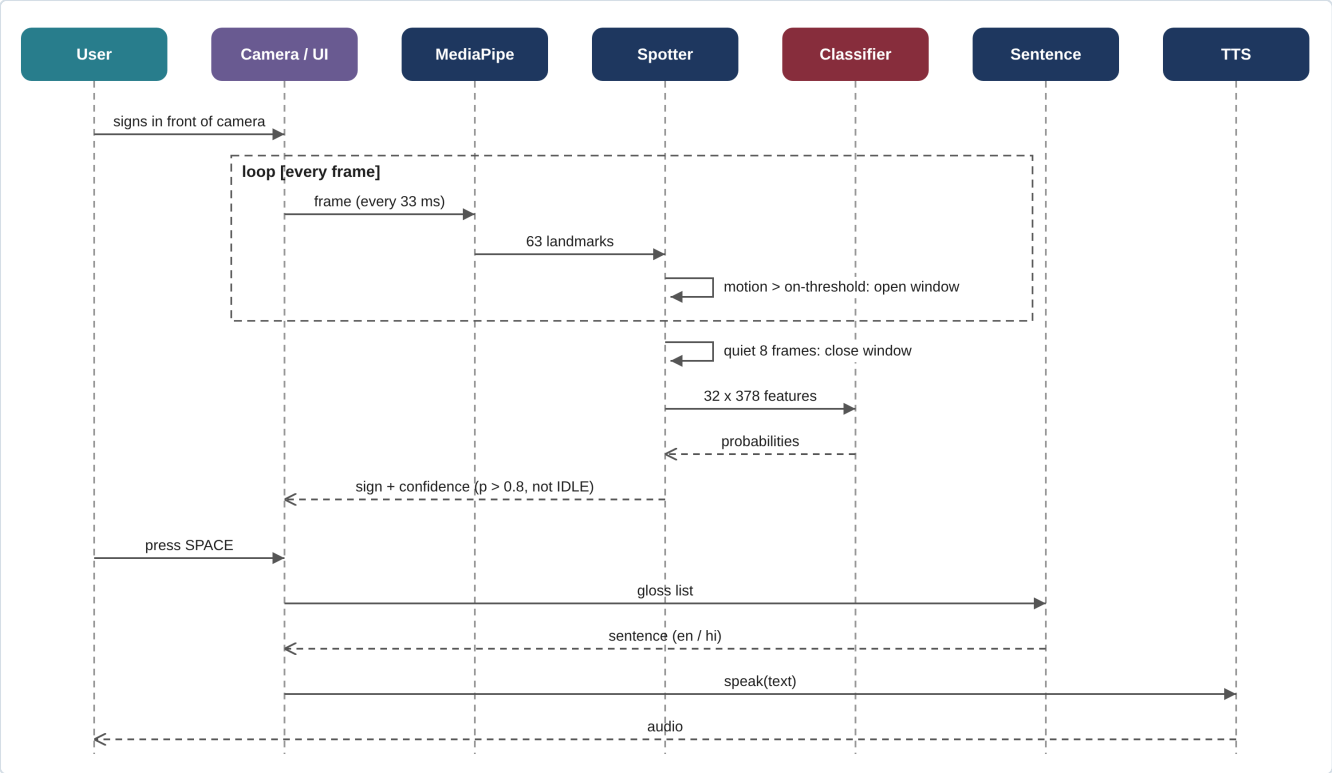
Layer	Output shape
Input	32 x 378
Linear 378 -> 256 + LayerNorm + GELU + Dropout 0.2	32 x 256
2-layer bidirectional GRU, hidden 128, dropout 0.3	32 x 256
Attention pooling: a(t) = softmax(w . h(t)), z = sum of a(t) h(t)	256
Dropout + Linear 256 -> 16	16 class scores (15 signs + IDLE)

- About **0.69 M parameters** (694,801), small enough for real-time CPU inference.
- Comparison models with the same input: Transformer encoder (4 layers, 4 heads, width 192) and a temporal 1D-CNN (3 convolution layers, 192 channels).

4. Sign spotter (real-time segmentation)

- Motion energy = largest wrist movement between frames, divided by shoulder width.
- Window opens when energy > 0.02; closes after energy < 0.01 for 8 frames (min 10, max 90 frames).
- A sign is accepted only if confidence >= 0.8 and the class is not IDLE; 12-frame cool-down stops duplicates.

Figure: `docs/diagrams/sequence.png` (one sign, then speak). If the slide gets crowded, put the sequence diagram on its own slide, "Detailed Design (contd.)".



sequence.png

SPEAKER NOTES

"Each frame becomes a 378-number vector. Dividing by shoulder width means a tall person and a short person, near or far from the camera, give the same numbers. The GRU reads the 32 frames in both directions, and attention lets the model focus on the frames where the hand shape is clearest."

Slide 12: Hardware and Software Requirements

ON THE SLIDE

HARDWARE

Item	Minimum	Used by us
Processor	Intel Core i5 8th gen / AMD Ryzen 5 (no GPU needed)	[XX]
RAM	8 GB	[XX]
Camera	720p front-facing webcam	[XX]
Audio	Speakers or headphones	[XX]

SOFTWARE

Item	Version / tool
Operating system	Windows 10/11, Ubuntu 22.04 or macOS 12+
Language	Python 3.10 or 3.11
Landmark detection	MediaPipe Holistic 0.10.14
Video and UI	OpenCV 4.10
Deep learning	PyTorch (CPU build) for training
Inference	ONNX, ONNX Runtime (CPU)
Data and evaluation	NumPy, scikit-learn, Matplotlib
Speech	pyttsx3 (offline English), gTTS + pygame (Hindi)
Testing and tools	pytest, Git and GitHub, draw.io

Training: a laptop CPU is enough for 800 clips (a few minutes per model); Google Colab is optional.

SPEAKER NOTES

"Everything is open source and runs on a normal student laptop. Internet is needed only to generate Hindi speech the first time; after caching, the whole system works offline."

Slide 13: Functionalities of the modules

ON THE SLIDE (table)

Module (file)	Input -> output	What it does
`record.py`	Webcam -> keypoint clips (.npy)	Guided recorder: shows the sign, counts down, records about 2.5 s, checks quality, retries; saves keypoints only, never video
`check_data.py`, `qc.py`	Recorded clips -> report	Checks layout and hand tracking; removes weak takes, flags possible wrong signs, lists re-recordings
`landmarks.py`	MediaPipe result -> (63, 3) per frame; clip -> 32 x 378	Selects 63 points; fills gaps; resamples to 32 frames; shoulder normalisation; velocity
`augment.py`	Clip -> varied clip	Speed change, rotation, scaling, shift, noise and hand drop-out, so the model sees more variety
`models.py`, `train.py`	Dataset -> trained model	BiGRU (main), Transformer and 1D-CNN; signer-independent and random splits; class-balanced loss
`evaluate.py`, `confusion.py`	Model + data -> metrics	Top-1, top-3, macro-F1, per-sign recall, confusion matrices and most-confused pairs
`export_onnx.py`, `inference.py`	PyTorch model -> ONNX	Export and check that ONNX output equals PyTorch output; fast CPU inference
`spotter.py`	Frame stream -> (sign, confidence)	Detects sign start and end from wrist motion; rejects IDLE and low confidence
`sentence.py`, `rules.json`	Gloss list -> English / Hindi sentence	Matches ISL patterns such as [PRONOUN, BODY, PAIN]; word-by-word fallback
`speech.py`	Text + language -> audio	pyttsx3 for English, gTTS for Hindi with a cache, speaks one sentence at a time
`app.py`	Webcam -> live UI and speech	Skeleton, status, FPS, top-3 confidence bars, signs, sentence; keys SPACE, u, c, l, s, f, q

SPEAKER NOTES

"The code is a modular Python package. Each module has one job and its own unit tests, and the vocabulary and sentence rules live in JSON files, so adding a sign does not need code changes."

Slide 14: Implementation

Split into two slides if it does not fit.

ON THE SLIDE

A. Data collection and quality control

- Guided recorder: 16 classes x 10 takes per signer, about 2.5 s per take, at 30 FPS.
- IDLE takes cover natural non-sign movements: resting hands, touching the face, adjusting hair, looking at a phone, shrugging.
- Only landmark coordinates are saved (`data/keypoints/<SIGN>/<signer>_<take>.npy`); signers are anonymous IDs (s01, s02, ...); written consent from every signer.
- Quality control rejects takes where a hand is tracked in fewer than 60% of frames, flags takes whose movement differs from the same signer's other takes (possibly a wrong sign), and lists every sign with fewer than 8 good takes for re-recording. Rejected clips are moved and logged, never deleted.
- Public INCLUDE-50 dataset (958 videos, 50 signs) converted to the same keypoint format, used to check the pipeline on real signing.

B. Training

- Augmentation: speed 0.8-1.2x, rotation ±15°, scale 0.9-1.1, shift, coordinate noise, hand drop-out.
- Optimiser AdamW (learning rate 0.001, weight decay 0.01) with cosine learning-rate decay.
- Cross-entropy with label smoothing 0.1; signs with fewer clips get a larger loss weight.
- Up to 120 epochs, batch size 64, early stopping after 25 epochs without improvement.
- Training clips are cut to the same window the live spotter would see, so training matches live use.

C. Evaluation protocol

- Signer-dependent: random split of clips (same people in training and test).
- Signer-independent: leave-one-signer-out (LOSO); train on 4 signers, test on the 5th, repeat 5 times.
- Final check on a 6th person (s06) recorded only for testing.

D. Deployment and live app

- Final model exported to ONNX; the export fails if ONNX and PyTorch outputs differ by more than 0.0001.
- Live app shows the skeleton, READY / SIGNING status, FPS, top-3 confidence bars, the signs and the sentence.
- Keys: SPACE speak, `u` undo, `c` clear, `l` English/Hindi, `s` skeleton, `f` full screen, `q` quit; optional auto-speak after a 2.5 s pause.

E. Sentence generation example

Signed (ISL order)	Pattern matched	English	Hindi
ME STOMACH PAIN	[PRONOUN, BODY, PAIN]	My stomach hurts.	मेरे पेट में दर्द है।
ME NEED WATER	[PRONOUN, NEED, THING]	I need water.	मुझे पानी चाहिए।
DOCTOR CALL	[PERSON, CALL]	Please call the doctor.	कृपया डॉक्टर को बुलाइए।

Figure (optional): a screenshot of the live app with the skeleton and a sentence on screen (only with the consent of the person shown).

SPEAKER NOTES

"We built our own dataset because no public ISL dataset has these hospital signs recorded with an IDLE class. Leave-one-signer-out is the key part of our evaluation: the test person is never in the training data, which is exactly the situation in a real hospital."

Slide 15: Results

Split into two slides: 15a (what is already verified) and 15b (measured accuracy and speed).

15a. Sentence generation (verified)

ON THE SLIDE

ID	Signed (ISL order)	English output	Hindi output
S1	ME HEAD PAIN	My head hurts.	मेरे सिर में दर्द है।
S2	ME STOMACH PAIN	My stomach hurts.	मेरे पेट में दर्द है।
S3	ME FEVER	I have a fever.	मुझे बुखार है।
S4	ME NEED WATER	I need water.	मुझे पानी चाहिए।
S5	ME NEED MEDICINE	I need medicine.	मुझे दवा चाहिए।
S6	ME NEED TOILET	I need to use the toilet.	मुझे शौचालय जाना है।
S7	TOILET WHERE	Where is the toilet?	शौचालय कहाँ है?
S8	DOCTOR CALL	Please call the doctor.	कृपया डॉक्टर को बुलाइए।
S9	HELP	Please help me.	कृपया मेरी मदद कीजिए।
S10	YES / NO	Yes. / No.	हाँ। / नहीं।

- The sentence engine produces all 11 expected sentences exactly: **11/11 in English and 11/11 in Hindi** (automated tests `tests/test_sentence.py` and `tests/test_app.py`).
- Classifier size: **about 0.69 M parameters**.
- 67 automated unit tests cover landmarks, spotting, training, sentences, speech, recorder and app. Run `python -m pytest -q` on your laptop and quote the pass count.

15b. Accuracy and real-time performance (fill in from your runs)

ON THE SLIDE

Metric	Target	Measured
Top-1 accuracy, signer-dependent (random split)	>= 95%	[XX]%
Top-1 accuracy, signer-independent (LOSO mean ± std)	>= 80%	[XX] ± [XX]%
Unseen person s06, live (TC-12)	>= 42 / 60	[XX] / 60
False triggers while idle (TC-10)	<= 1 per minute	[XX]
Frame rate on the weakest laptop (TC-01)	>= 15 FPS	[XX] FPS
Hands stop -> text on screen (TC-14)	<= 500 ms	[XX] ms
Sentence correctness S1-S10 (TC-07, TC-08)	10 / 10	[XX] / 10 live; 11 / 11 automated

Model comparison (Table R2), if you have time for a third results slide:

Model	Random split top-1	LOSO top-1	LOSO macro-F1	Parameters	CPU ms / sign
BiGRU + attention	[XX]	[XX] ± [XX]	[XX]	0.69 M	[XX]
Transformer	[XX]	[XX] ± [XX]	[XX]	[XX]	[XX]
1D-CNN	[XX]	[XX] ± [XX]	[XX]	[XX]	[XX]

Also show the confusion matrix image from `signvaani.confusion` and name the two most confused sign pairs with the reason (same hand shape, different location or movement).

Where each number comes from:

Number	Command
Random split	`python -m signvaani.train --data data/keypoints --random-split`
LOSO	`python -m signvaani.train --data data/keypoints --loso`
Model comparison (R2)	`python -m signvaani.experiments compare --out docs/results/model_comparison.md`
Ablation (R4)	`python -m signvaani.experiments ablate --model gru --out docs/results/ablation.md`
Confusion matrix (R3, R6)	`python -m signvaani.confusion runs/loso_gru_<time>.json`
Unseen signer s06	`python -m signvaani.evaluate --ckpt models/best.pt --data data/unseen --signers s06 --out docs/results/s06`
FPS and latency	`python -m signvaani.app --model models/sign_model.onnx --log runs/live_log.csv`, then read the lines printed on `q`

SPEAKER NOTES

"Accuracy on the random split is higher than on unseen signers. That gap is a finding, not a failure: different people have different hand sizes, speeds and styles. Leave-one-signer-out is the number that matters for a real hospital, because every patient is a new person."

Slide 16: Applications

ON THE SLIDE (six short blocks, each with an icon)

1. **Hospitals and clinics:** patients tell doctors and nurses about pain, fever, medicine, water or the toilet without waiting for an interpreter.
2. **Public service counters:** banks, railway stations, post offices and government offices, where staff do not know ISL.
3. **Emergencies and help desks:** quick requests such as HELP or CALL DOCTOR, spoken aloud at once.
4. **Education:** deaf students and ISL learners practise signs and get instant feedback; hearing classmates and teachers can follow.
5. **Low-cost and offline settings:** runs on an ordinary laptop CPU with a webcam, with no internet for English, so it suits rural health centres.
6. **Research:** a privacy-preserving keypoint dataset and a signer-independent evaluation protocol that others can reuse and extend to new domains.

SPEAKER NOTES

"The same pipeline works for any domain: only the vocabulary file and the sentence rules change. The tool supports human interpreters; it does not replace them in medical, legal or emergency decisions."

Optional extra slide after slide 16: Conclusion and Future Scope

ON THE SLIDE

CONCLUSION

- Built SignVaani, a complete webcam-to-speech pipeline for ISL that runs in real time on a laptop CPU.
- A compact representation of 63 hand, body and face landmarks with shoulder normalisation and velocity.
- Automatic sign spotting with an IDLE class, so the user signs naturally without pressing a button per sign.
- An ISL-order sentence engine produces correct English and Hindi sentences for all demo sentences.
- Honest, signer-independent evaluation, and a privacy-first design: no video is ever stored.

LIMITATIONS

- Isolated signs from one domain (15 signs + IDLE); a small number of signers.
- No full ISL grammar (spatial referencing, classifiers) and no fingerspelling.
- Hindi speech needs internet unless cached in advance.

FUTURE SCOPE

- Larger vocabulary and continuous sentence-level ISL translation, trained on datasets such as iSign [4].
- More Indian languages (Telugu, Kannada, Tamil) with IndicTrans2 [20] and offline Indic-TTS [21].
- Graph-based models such as ST-GCN [16] for better accuracy on similar signs.
- Android / mobile app and a reverse speech-to-sign avatar for two-way conversation.
- User-specific fine-tuning on a few samples from each new user.

Slide 17: References

Follow the template's format. Split over two or three slides if needed.

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Slide 18: Thank You

ON THE SLIDE

- THANK YOU
- (Optional, smaller) Questions?

SPEAKER NOTES

Be ready to show the live demo (3 minutes): sit still for 15 s to show no false signs, sign ME STOMACH PAIN and press SPACE, press `l` and sign ME NEED WATER in Hindi, then have a teammate who was not in the training data sign DOCTOR CALL. If the camera or room fails, play the recorded demo video.